

PART A

PUBLICATIONS

Title	
User Churn Prediction within Music Streaming Service Industry: A Comparison of Machine Learning Models	
Havard Reference	Matusevičius, M. (2021) <i>User churn prediction within music streaming service industry: A comparison of machine learning models</i> . MSc thesis. Tilburg University.
Link	https://arno.uvt.nl/show.cgi?fid=157849
Modelling	• Logistic Regression • Decision Tree • Random Forest • Extreme Gradient Boosting (XGBoost) • LR, DT and RF were treated as baseline/simple models. • XGBoost was treated as the more complex model and was tuned using random search over 100 configurations. • The models used a 75% training, 10% validation and 15% testing split. • Evaluation included accuracy, precision, recall and Cohen's kappa
Preprocessing techniques	• removed missing user-log records; • aggregated daily user logs to the user level; • converted daily listening behaviour into monthly listening-pattern features; • removed the raw date after aggregation; • created days_listened; • merged datasets using the user ID; • filtered implausible ages outside 5 to 99; • removed remaining missing observations; • removed registration_init_time; • addressed class imbalance using NearMiss-1 undersampling; • one-hot encoded categorical variables
Datasets used	• KKBox train • KKBox members • KKBox user_logs
Key takeaways	1. Behavioural aggregation is essential. Raw daily user logs need to be converted into customer-level behavioural features before modelling. 2. Class imbalance must be explicitly addressed. The study found approximately 12.7% churn after its preprocessing and used NearMiss-1 undersampling. 3. A baseline-to-complex model comparison is useful. The study provides a rationale for comparing interpretable baseline models against boosted/tree-based models rather than using only one algorithm.

Title	Beyond Accuracy: Benchmarking Machine Learning Models for Efficient and Sustainable SaaS Decision Support
Havard Reference	Mavridou, E., Vrochidou, E., Selvesakis, M. and Papakostas, G.A. (2025) 'Beyond Accuracy: Benchmarking Machine Learning Models for Efficient and Sustainable SaaS Decision Support', <i>Future Internet</i> , 17(10), 467. doi: 10.3390/fi17100467.
Link	https://www.mdpi.com/1999-5903/17/10/467
Modelling	• Logistic Regression • Decision Tree • Random Forest • XGBoost • LightGBM • CatBoost • MLP • Naïve Bayes variants • Bagging • SGD • XGBoost + LightGBM voting ensemble • other voting/stacking ensembles.
Preprocessing techniques	• missing-value handling; • categorical-variable encoding; • feature scaling; • stratified 80/20 train-holdout split; • 10-fold cross-validation on the training portion; • sample weighting using balanced class weights to address class imbalance.
Datasets used	• KKBOX churn prediction dataset • Telco Customer Churn dataset
Key takeaways	1. Do not rely on accuracy alone. The KKBOX dataset is imbalanced, so recall, precision, F1, AUC and especially probability-based measures such as log loss provide additional information. 2. Tree-based boosting models are highly relevant candidates. XGBoost and LightGBM were included in the benchmark and performed substantially differently from simpler models on KKBOX. 3. A consistent experimental setup matters. Using the same train/test partitions and evaluation procedure across models makes model comparison more defensible.

Title	Advanced customer churn prediction for a music streaming digital marketing service using attention graph-based deep learning approach
Havard Reference	Cheng, H. and He, J. (2025) 'Advanced customer churn prediction for a music streaming digital marketing service using attention graph-based deep learning approach', <i>Scientific Reports</i> , 15, 45028. doi: 10.1038/s41598-025-28357-z.
Link	https://www.nature.com/articles/s41598-025-28357-z
Modelling	• Graph Attention Network for user-to-user relational information; • MLP for tabular customer features; • four attention heads; • dropout; • batch normalization; • ReLU activations; • weighted binary cross-entropy; • Adam optimisation; • early stopping based on validation AUC.
Preprocessing techniques	• aggregation of transactions by user; • calculation of total transactions; • calculation of cumulative payment; • calculation of cancellation counts; • extraction of the most recent transaction date; • processing of over 30 GB user logs in chunks; • calculation of active days; • calculation of total listening seconds; • calculation of total unique songs played; • merged behavioural, transaction, demographic and target data at user level; • created membership duration; • extracted registration year/month; • label encoded categorical variables; • log transformed heavily skewed listening variables; • imputed missing values with zero; • standardised numerical variables using Z-score scaling.
Datasets used	The four KKBox tables: • train_v2.csv • transactions_v2.csv • user_logs_v2.csv • members_v3.csv
Key takeaways	1. Behavioural, transaction and demographic information can be combined into customer-level features. 2. The severe class imbalance should be handled explicitly without automatically destroying the original population distribution. 3. More complex deep-learning approaches exist, but they introduce substantially greater modelling complexity.

PUBLIC DATASET

Name	WSDM - KKBox's Churn Prediction Challenge
Havard Reference	KKBOX (2017) <i>WSDM – KKBox's Churn Prediction Challenge</i> . Kaggle. Available at: https://www.kaggle.com/competitions/kkbox-churn-prediction-challenge (Accessed: 20 September 2026).
Link	https://www.kaggle.com/competitions/kkbox-churn-prediction-challenge
Number of instances	KKBox is a relational dataset containing multiple tables. The key labelled population consists of 970 960 users in train_v2.csv, where each observation contains a user identifier (msno) and a binary churn target (is_churn). To complement this main dataset, the supporting tables capture data at different observation grains: • members_v3.csv is organized at the customer level • transactions_v2.csv tracks individual transactions • user_logs_v2.csv records activity at a user-day level.
Number of feature columns	Across the four tables, the raw schema contains a total of 21 predictor fields (excluding the user identifier msno and target is_churn). members_v3.csv = 5 predictor fields transactions_v2.csv = 8 predictor fields user_logs_v2.csv = 8 predictor fields train_v2.csv = 0 predictor fields with 1 target field However, this raw count does not represent the final modelling feature total. Because the transaction and user-log tables maintain a one-to-many relationship with each customer, these records must be aggregated to the customer level before modelling.
Number of classes (if applicable)	This is a binary classification problem where the target variable is_churn indicates whether a customer renewed (0) or churned (1). Per KKBox's definition, a customer is classified as churned if they fail to resubscribe within 30 days after their previous subscription expires. The literatures above indicate a notable class imbalance in the original labelled population, consisting of approximately 91% non-churn and 9% churn instances.
Target range (if applicable)	Not applicable
Origin	The dataset is a real-world, anonymised observational dataset rather than a synthetically generated one. It is provided by KKBOX, which is an Asian music-streaming service, for the WSDM 2018 Churn Prediction Challenge hosted through Kaggle. It was designed specifically to predict subscription-user churn and contains customer and member information, subscription and transaction history, daily listening behaviour, and churn labels.